Delhi Public School

Subject: Physics | Grade: 11th | Class: 11th

Time: 2 Hours | Maximum Marks: 20

Name: ________________________ Roll No: ___________ Section: ___________

General Instructions:

1. *Read all questions carefully before answering.*
2. *Write neatly and legibly.*

Section A

Choose the correct option for each of the following multiple-choice questions. Each question carries 1 mark.

Q1. The inertia of an object is directly dependent on and proportional to which of the following physical quantities? [Easy] **[1]**

a. Velocity
b. Acceleration
c. Mass
d. Force

Q2. When a stationary bus suddenly starts moving forward, the passengers standing inside tend to fall backward. This phenomenon is a direct consequence of: [Easy] **[1]**

a. Inertia of rest
b. Inertia of motion
c. Inertia of direction
d. Conservation of momentum

Q3. A constant force of 10 N acts on a body of mass 2 kg for a duration of 3 seconds. The magnitude of the change in momentum of the body is: [Easy] **[1]**

a. 5 kg m/s
b. 15 kg m/s
c. 30 kg m/s
d. 60 kg m/s

Q4. A block of mass 5 kg is sliding on a rough horizontal surface with a coefficient of kinetic friction of 0.2. What is the magnitude of the kinetic frictional force opposing its motion? (Take $g = 10\ m/s^2$) [Moderate] **[1]**

a. 2 N
b. 5 N
c. 10 N
d. 50 N

Q5. A particle of mass m moves in a circular path of radius r with a constant speed v under the action of a centripetal force. The total work done by this centripetal force during a semi-circular displacement is: [Hard] **[1]**

a. $\pi m v^2$
b. $m v^2 / r$
c. Zero
d. $2 m v^2$

Section B

Answer the following short answer questions briefly. Each question carries 2 marks.

Q6. State Newton's First Law of Motion and explain how it provides a qualitative definition of force. [Easy] **[2]**

Q7. Using the concept of resolution of forces, explain why pulling a lawn mower is easier than pushing it. [Moderate] **[2]**

Q8. A vehicle of mass 1000 kg is moving with a velocity of 20 m/s. If it is brought to a complete stop in 4 seconds by applying brakes, calculate the average retarding force exerted on the vehicle. [Easy] **[2]**

Q9. Distinguish between static friction, limiting friction, and kinetic friction. Draw a simplified graph showing how frictional force varies with the applied force. [Moderate] **[2]**

Q10. A bullet of mass 20 g moving with a speed of 150 m/s penetrates a sandbag and comes to rest in 0.03 seconds. Calculate the depth of penetration of the bullet into the sandbag. [Moderate] **[2]**

Section C

Answer the following long answer questions. Show detailed derivations and calculations where applicable. Each question carries 5 marks.

Q11. (a) State the law of conservation of linear momentum and prove it using Newton's Third Law of Motion. (b) A shell of mass 0.02 kg is fired from a gun of mass 100 kg. If the muzzle speed of the shell is 80 m/s, calculate the recoil speed of the gun. [Moderate] **[5]**

Q12. Derive an analytical expression for the maximum safe speed of a vehicle negotiating: (a) A flat horizontal circular road with a coefficient of static friction μ. (b) A circular road banked at an angle θ, neglecting friction. [Moderate] **[5]**

Q13. (a) Define the coefficient of static friction and the angle of friction. Prove mathematically that the coefficient of static friction is equal to the tangent of the angle of friction. (b) A block of mass 4 kg rests on an inclined plane. When the angle of inclination of the plane is gradually increased to 30°, the block just begins to slide. Determine the coefficient of static friction between the block and the plane. [Hard] **[5]**

Q14. (a) Define the term 'Impulse'. Establish the Impulse-Momentum Theorem. (b) A batsman hits a cricket ball of mass 0.15 kg straight back in the direction of the bowler without changing its initial speed of 12 m/s. Assuming the motion of the ball is linear, calculate the magnitude of the impulse imparted to the ball. [Moderate] **[5]**

Q15. Design a complete free-body diagram analysis for a system consisting of two unequal masses m_1 and m_2 (where $m_1 > m_2$) connected by a light inextensible string passing over a smooth, frictionless pulley (Atwood's machine). Derive general expressions for: (a) The common acceleration of the system. (b) The tension in the string. [Hard] **[5]**

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